

# Muhua (黄牧华) Huang

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I study how AI systems fundamentally differ from **human intelligence** and how they, as **social actors**, reshape **collaboration** and **collective decision-making**. My research combines **computational social science**, **psychometrics**, and **generative agent-based modelling (GABM)** to investigate AI's unique cognitive characteristics, uncover new insights about human behavior from AI representations, and understand how AI agents alter **interaction dynamics** within **groups and organizations**.

**Research:** Human-AI Collaboration, Organizational Behavior, AI Alignment, Computational Social Science  
**Engineering:** GABM, Large-Scale LLM Experimentation, Fine-tuning, Agent, High Performance Computing  
**Analytical:** Psychometrics, Embedding Models, Social Network Analysis, ML/NLP, Behavioral Experiment

## EDUCATION

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|-----------------------------------------------------------------|-----------------|
| <b>Stanford Graduate School of Business</b>                     | 2025.9 – 2030.6 |
| PhD in Organizational Behavior                                  |                 |
| <b>The University of Chicago</b>                                | 2023.9 – 2025.6 |
| MA in Computational Social Science                              |                 |
| <b>The University of British Columbia (UBC)</b>                 | 2018.9 – 2023.5 |
| BA in Computer Science & Psychology (Honours)                   |                 |
| <b>International AI Evaluation Programme Fellow</b>             | 2026.1 – 2026.5 |
| <b>Summer Institute in Computational Social Science (SICSS)</b> | 2024.6 – 2024.7 |
| <b>Google Research Mentorship Program Scholar</b>               | 2020.9 – 2021.1 |

## PUBLICATIONS

### AI Simulation

- [1] **Huang, M.**, & Evans, J. (2026). Institutions as cached computation for resource-rational negotiation. *Behavioral and Brain Sciences*, 49, e144. [DOI](#)
- [2] **Huang, M.**, Zhang, X., Soto, C., & Evans, J. (2026). Designing AI-Agents with Personalities: A Psychometric Approach. *Personality Science*. [DOI](#)
- [3] Bai, Y., Duan, S., **Huang, M.**, Yao, J., Liu, Z., Zhang, P., ...& Xie, X. (2026). IROTE: Human-like Traits Elicitation of Large Language Model via In-Context Self-Reflective Optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*. [DOI](#)

### AI Character and Value Alignment

- [1] Yao, J., Yi, X., Duan, S., Wang, J., Bai, Y., **Huang, M.**, ...& Xie, X. (2025). Value compass benchmarks: A comprehensive, generative and self-evolving platform for LLMs' value evaluation. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. [DOI](#)
- [2] Kim, H., Yi, X., Bak, J., Yao, J., Lian, J., **Huang, M.**, Duan, S., & Xie, X. (2025). The Road to Artificial SuperIntelligence: A Comprehensive Survey of Superalignment. *SuperIntelligence - Robotics - Safety & Alignment*, 2(1). [DOI](#)

## Social Science

- [1] Savalei, V., & **Huang, M.** (2025). Fit Indices Are Insensitive to Multiple Minor Violations of Perfect Simple Structure in Confirmatory Factor Analysis. *Psychological Methods*. DOI
- [2] Zhang, X., **Huang, M.**, Sun, J., & Savalei, V. (2025). Improving the Measurement of the Big Five via Alternative Formats for the BFI-2. *Journal of Personality Assessment*. DOI
- [3] Laurin, K., Engstrom, H., & **Huang, M.** (2024). What will my life be like when I'm 25? How social class predicts kids' answers to this question, and how their answers predict their futures. *Journal of Social Issues*. DOI

## INDUSTRY EXPERIENCE

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**Student Researcher** | Google DeepMind 2026.6 – 2026.9

- Led research on which AI-agent skills transfer across occupations, separating general work skills from occupation-specific capabilities.
- Building a skill taxonomy from large-scale labor-market data and an evaluation framework designed to predict agent performance on occupations not used in development.
- Lead-authored a *Harvard Business Review* perspective (in preparation) with Google DeepMind's Head of Organizational AI Research, arguing that training on aggregate data and consensus preferences pulls LLMs toward convergent ideas, while taste grounded in local, individual experience matters in early-stage innovation.
- Synthesized research on model homogenization, human creativity, and decision-making under uncertainty into strategies for leaders to use AI without flattening their judgment.

**Contributing Researcher and Writer** | Stanford HAI & Google DeepMind 2026

- Partnered with Martin Gonzalez on conceptualization, research, and writing for *Organizing Intelligence*, a 16-topic research primer on AI and organizations; publicly acknowledged in the report.

**Research Intern** | Microsoft Research Asia 2024.7 – 2024.10

Advisor: Dr. Xiaoyuan Yi and Dr. Xing Xie

**On the Dynamics of LLMs' Values as a Community**

- Investigated AI value alignment and the unique structure of AI systems' values; compared strategies for eliciting personality traits in LLMs.
- Built scalable multi-agent simulations with cognitive modules, RAG, and parallel processing to examine value diversity and emergence in AI-agent communities.
- Designed experiments varying value diversity, population size, and time; used social network analysis, GraphRAG, and NLP to study network formation and emergent language, culture, and institutions.

## WORKING PAPERS

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- [1] **Huang, M.**, & Guilbeault, D. (In Prep). AI as Alien Intelligence: Shared Vocabulary Masks Divergent Category Systems in Human and AI Collectives. *Manuscript available upon request*.

## CONFERENCE PRESENTATIONS

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† first author / \* solo

- \* **Designing AI-Agents with Personalities: A Psychometric Approach**  
APA, Seattle (Aug 2024, Poster); CPA, St. John's (Jun 2025, w/ X. Zhang); CPA, Montréal (Jun 2026, w/ X. Zhang); ISDSA, Beijing (Jul 2026, Speed Talk, w/ X. Zhang)
- † **Beyond Anthropomorphism: Unveiling Unique Value Structure of Large Language Models**  
(w/ P. Biedma, X. Yi, L. Huang, M. Sun, J. Evans, X. Xie)  
IC2S2, Norrköping (Jul 2025)

- <sup>†</sup> **Re-Discovering the Big Five: Using LLM Embeddings to Understand Personality Structure** (w/ N. Huang)  
APA, Denver (Aug 2025)
- <sup>†</sup> **Improving the Measurement of the Big Five via Alternative Formats for the BFI-2** (w/ X. Zhang, V. Savalei)  
APA, Washington DC (Aug 2023, Poster); *IMPS*, Minneapolis (Jul 2025, w/ J. Sun); *CPA*, Montréal (Jun 2026)
- <sup>†</sup> **Social Class Differences in Children’s Hopes and Expectations for the Future** (w/ H. Engstrom, K. Laurin)  
APS (May 2021, Poster); *SPSP*, San Francisco (Feb 2022, Talk); *Society in the Classroom*, London (Jul 2022, Talk)
- **Understanding Silicon Sampling through a Psychometric Lens** (w/ C. F. Falk, X. Zhang, N. Guenole, W. Li)  
CPA, Montréal (Jun 2026)
- **Can SEM Fit Indices Distinguish Between CFA and EFA Data Structures?** (w/ V. Savalei)  
SMEP, Iowa City (Oct 2023)
- **Gauging Student Engagement with an XAI Interface via Eye-tracking** (w/ C. Conati, R. Murali)  
IJCAI Workshop on XAI (Jun 2022)

## INVITED TALKS

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- [1] Yang, L., Sun, L., **Huang, M.**, Wang, J., Jiang, R., & Xu, F. (November 14, 2024). Panel discussion: LLM-driven social science and generative agents. *2024 MSR Asia TAB Workshop: Shaping the Future with Societal AI*. Microsoft Research Asia, Beijing, China.
- [2] **Huang, M.** (January 13, 2025). Designing LLM-agents with personalities: A psychometric approach. Invited talk at the *Quantitative Methods Forum*, York University, Toronto, Canada.

## RESEARCH EXPERIENCE

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**PhD Student Researcher** | Stanford Graduate School of Business 2025.10 – Present  
Advisor: Dr. Douglas Guilbeault

### AI as Alien Intelligence: Shared Vocabulary Masks Divergent Category Systems in Human and AI Collectives

- Engineered a large-scale multi-agent coordination experiment (~375,000 AI trials with GPT-4o agents in network structures of  $N=2$  to 50), replicating a human category-formation paradigm.
- Discovered “representational relativity”: despite comparable behavioral performance, human and AI collectives resolve ambiguity in systematically different places ( $r = -0.130$ ,  $p < 0.001$ ).
- Showed AI labels are disproportionately visual-dominant and concrete, while human labels draw on richer multimodal experience — divergence driven by grounding modality, not training data.
- Found that AI agents strategically invent novel vocabulary to express distinctions human language cannot; forcing AI to use only human labels reduces performance by 12.7%.

**Master’s Thesis** | University of Chicago 2023.9 – 2024.6  
Advisor: Dr. James Evans

### Designing LLM-Agents with Personalities: A Psychometric Approach

- Developed the first psychometrically validated framework for assigning Big Five personalities to LLM agents, with controllable, fine-grained design across multiple models (GPT-3.5, GPT-4, Claude 3).
- Validated that personality-assigned agents exhibit predictable behavioral differences in moral reasoning, risk-taking, and cooperation (personality explains 38–52% of variance in agent decisions).
- Collected data from 350 human participants for direct human–agent comparison, demonstrating potential for using personality-designed agents in behavioral research.
- Published in *Personality Science*; methodology enables designing AI agents for specific collaborative roles.

**Research Assistant, Quantitative Psychology** | University of British Columbia 2020.4 – 2023.8

Advisor: Dr. Victoria Savalei

### Exploring the Factor Structure of the BFI-2 in Alternative Scale Formats

- Compared four personality scale formats using SEM; found that alternative formats (Expanded, Item-Specific) significantly improve reliability and model fit over Likert. Published in *Journal of Personality Assessment*.
- Honours thesis nominated for multiple Canadian national undergraduate research awards (Belkin, CPA).

### SEM Fit Indices' Sensitivity to Omitted Crossloadings in CFA

- Built an R Shiny simulation app to examine fit index performance when CFA models are applied to data with EFA-like structure; contributed to manuscript published in *Psychological Methods*.

**Research Assistant, Social Psychology** | University of British Columbia 2020.8 – 2021.9

Advisor: Dr. Kristin Laurin

### Socioeconomics and Imagined Future Study

- Applied ML (Random Forest, Neural Network) and NLP (Topic Modelling, LSTM, Word2Vec) to analyze 10,000+ youth essays, revealing how socioeconomic background shapes imagined futures. Published in *Journal of Social Issues*.
- Supervised 20 research assistants and built data processing protocols including a word distribution and performance tracking system.

**Research Assistant, Human-Centered AI** | University of British Columbia 2021.10 – 2022.9

Advisor: Dr. Christina Conati

### Personalized Explainable AI for Intelligent Tutoring System Study

- Evaluated how personalizing AI explanations to individual cognitive profiles (perceptual speed, personality) affects learning outcomes, using eye-tracking and linear mixed models.
- Found that tailoring AI tutoring interfaces to users' cognitive abilities improves engagement and effectiveness — demonstrating the value of adaptive human-AI interaction design.

## PROFESSIONAL EXPERIENCE

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**Computational Social Science Workshop Coordinator** | University of Chicago 2024.10 – 2025.6

- Organized the weekly MACSS Computational Social Science Workshop, featuring speakers on advanced computational methods and interdisciplinary research.

**Deep Learning Textbook Editor** | University of Chicago 2023.10 – 2024.10

- Proofread and edited “Thinking with Deep Learning” by Dr. James Evans; designed LaTeX templates for the textbook and for Journal of Social Computing.

**Teaching Assistant** | University of British Columbia 2021.01 – 2022.12

- TA for Human Computer Interaction Methods and Systematic Program Design; rated 4.7/5 by students and received three return offers.

**Crisis Respondent** | Kids Help Phone 2019.09 – 2020.08

- Provided over 230 hours of crisis counselling to 550+ individuals in severe mental distress.

## HONORS & AWARDS

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|--------------------------------------------------------------------------------------------------------------|------------------|
| Stanford Graduate School of Business Doctoral Fellowship                                                     | 2025 – 2030      |
| Stanford University EDGE Doctoral Fellowship (\$12,800)                                                      | 2025             |
| UChicago Outstanding Thesis Award (\$1,000)                                                                  | 2025             |
| Microsoft Research Asia Outstanding Intern Award (Stars of Tomorrow Certificate)                             | 2024             |
| OpenAI Researcher Access Program (\$3,500)                                                                   | 2024             |
| SICSS-Beijing Merit Based Scholarship (¥3,600)                                                               | 2024             |
| UChicago Computational Social Science Research Poster Competition (Honorable Mention, 2 <sup>nd</sup> Place) | 2024             |
| UChicago Quadrangle Research Scholarship (\$80,000)                                                          | 2023, 2024       |
| UChicago Social Sciences Promise & Merit Scholarship (\$10,000)                                              | 2023, 2024       |
| Quinn Research Assistantship Award (\$8,300)                                                                 | 2021             |
| UBC International Community Achievement Award (\$5,000)                                                      | 2021             |
| UBC Trek Excellence Scholarship (\$2,000)                                                                    | 2019, 2020       |
| UBC Faculty of Arts International Student Scholarship (\$8,800)                                              | 2019, 2020, 2022 |